

Multidataset-Health

Investigating Generalization in Health Risk Prediction
using Multi-Dataset Learning

Members

SSR+

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Problem Statement

Health prediction models trained on a single dataset **often fail to generalize to other datasets** due to differences in data distribution and feature representation.

What we do is **combine multiple datasets — using both Union and Intersection** feature strategies — to train shared models, then evaluate whether **these models can outperform single-dataset baselines when tested on individual datasets.**

Workflow

1. Data Preparation
2. Training Setup
3. Model Baseline
4. Concatenation Setup
5. Concatenation Result
6. Feature Importance
7. Error Analysis
8. Conclusion

1. Data Preparation

1. Sources

- **Statlog Heart Disease** - Dataset for Heart Disease prediction
- **Coronary Heart Disease** - CHD cases and clinical variables describing patient health conditions.
- **Framingham** - Dataset used to predict 10-year CHD risk.
- **Heart Disease** - Dataset used to predict Heart Disease
- **Stroke** - Dataset used to predict Stroke Risk

2. Dataset Columns

- **Statlog Heart Disease** - age, sex, chest-pain, rest-bp, serum-chol, fasting-blood-sugar, electrocardiographic, max-heart-rate, angina, oldpeak, slope, major-vessels, thal, heart-disease
- **Coronary Heart Disease** - sbp, tobacco, ldl, adiposity, famhist, typea, obesity, alcohol, age, chd

1. Data Preparation

2. Dataset Columns (cont.)

- **Framingham** - Sex, age, education, currentSmoker, cigsPerDay, BPMeds, prevalentStroke, prevalentHyp, diabetes, totChol, sysBP, diaBP, BMI, heartRate, glucose, TenYearCHD
- **Heart Disease** - age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal, target
- **Stroke** - id, gender, age, hypertension, heart_disease, ever_married, work_type, Residence_type, avg_glucose_level, bmi, smoking_status, stroke

3. Data Cleaning

- Rename Columns
- Change Data types
- Map Values (eg. .map() in pandas)
- Handle Missing Values
- Drop Duplicated Rows
- Feature Scaling

2. Training Setup

- **Train Test Split:** 80/20 (stratified)
- **Cross Validation** with StratifiedKFold (k=5) to preserves class distribution in each fold
- **Imbalanced handling** with SMOTE applied when class ratio ≥ 2.0

- **8 Classification Models**

- Logistic Regression
- KNN
- Decision Tree
- SVM
- Naive Bayes
- Random Forest
- XGBoost
- LightGBM

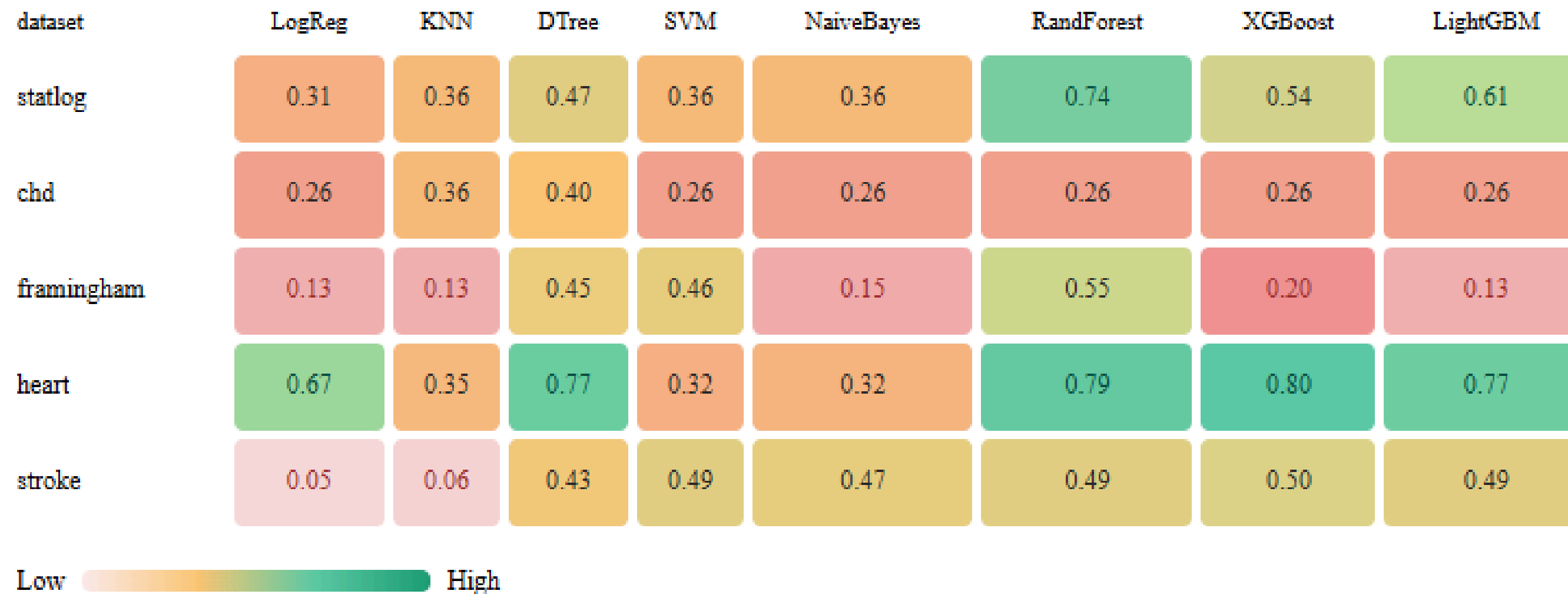
Model	Loss Function
Logistic Regression	Binary Cross-Entropy (Log Loss)
K-Nearest Neighbors	N/A (distance-based)
Decision Tree	Gini Impurity
SVM (RBF Kernel)	Hinge Loss
Naive Bayes	N/A (probabilistic / Bayes' theorem)
Random Forest	Gini Impurity
XGBoost	Binary Cross-Entropy (Log Loss)
LightGBM	Binary Cross-Entropy (Log Loss)

3. Model Baseline

- Each model is trained and tested on the same single dataset
- 5 datasets × 8 models = 40 baseline models
- Datasets: Statlog, CHD, Framingham, Heart, Stroke
- Models: Logistic Regression, KNN, Decision Tree, SVM, Naive Bayes, Random Forest, XGBoost, and LightGBM
- Evaluation metrics: Accuracy, F1 Weighted, F1 Macro, and Recall Minority
- Focus on **F1 Macro**

3. Model Baseline

F1 Macro

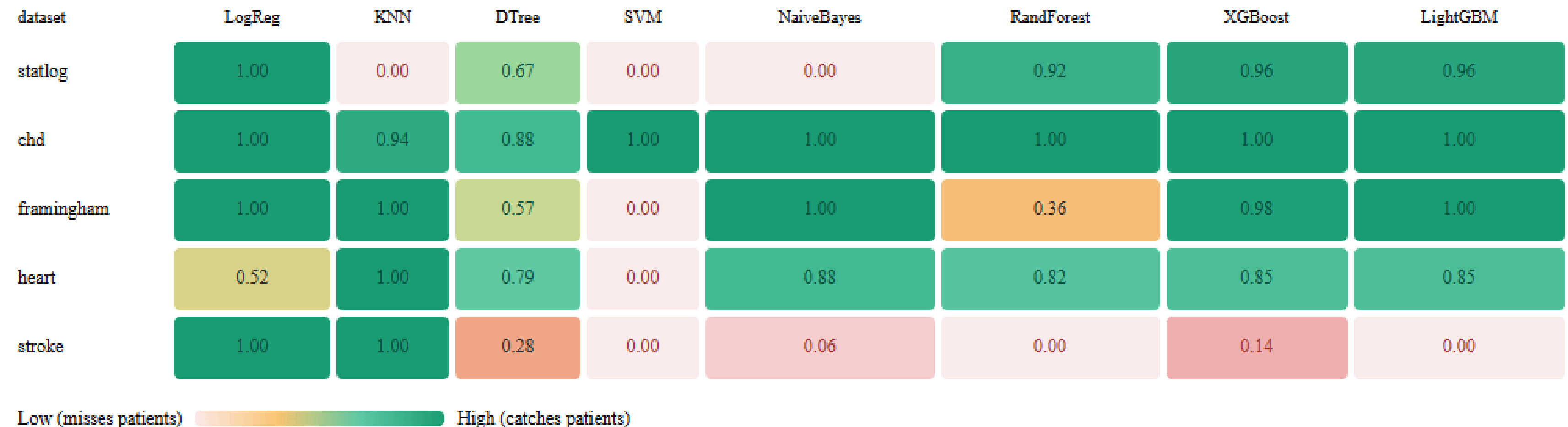


- **Ensemble models (RF, XGBoost, LightGBM) perform best** on **heart** and **statlog**
- **framingham**, **stroke**, and **chd** show generally **low F1 Macro** across most models
- Low F1 Macro on **stroke** and **framingham** reflects severe class imbalance

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3. Model Baseline

Recall Minority



- **SVM predicts negative for nearly all cases** → recall $\approx$ 0.00 across datasets
- **chd** shows **high recall but low F1 Macro** → models are biased toward the positive class
- **stroke ensemble models (RF, LightGBM) achieve recall = 0.00** — class imbalance effect

4. Concatenation Setup

1. Union

- Merge all features from all datasets
- Missing values
 - numeric: imputed with median
 - categorical: imputed with mode
- Scale with MinMaxScaler

2. Intersection

- Keep only shared features across datasets
- No missing values → scale with MinMaxScaler directly

Dataset Combinations

- 2-dataset combinations: $10 \text{ pairs} \times 2 \text{ strategies} = 20 \text{ configs}$
- 3-dataset combinations: $10 \text{ triplets} \times 2 \text{ strategies} = 20 \text{ configs}$
- Total: 40 concat configurations

4. Concatenation Setup

Rename Columns (Display only changed columns)

Statlog Heart Disease

- chest-pain → cp_type
- rest-bp → sbp
- serum-chol → chol
- fasting-blood-sugar → fbs
- electrocardiographic → ecg
- max-heart-rate → hr
- major-vessels → mv
- heart-disease → target

Coronary Heart Disease

- obesity → bmi
- chd → target

4. Concatenation Setup

Rename Columns (Display only changed columns)

Framingham

- Sex → sex
- education → edu
- currentSmoker → smoking_status
- cigsPerDay → num_cigs_per_day
- BPMeds → bp_status
- prevalentStroke → stroke_status
- prevalentHyp → hypertension_status
- diabetes → diabetes_status
- totChol → chol
- sysBP → sbp
- diaBP → dbp
- BMI → bmi
- heartRate → hr
- glucose → glucose_level
- TenYearCHD → target

4. Concatenation Setup

Rename Columns (Display only changed columns)

Heart Disease

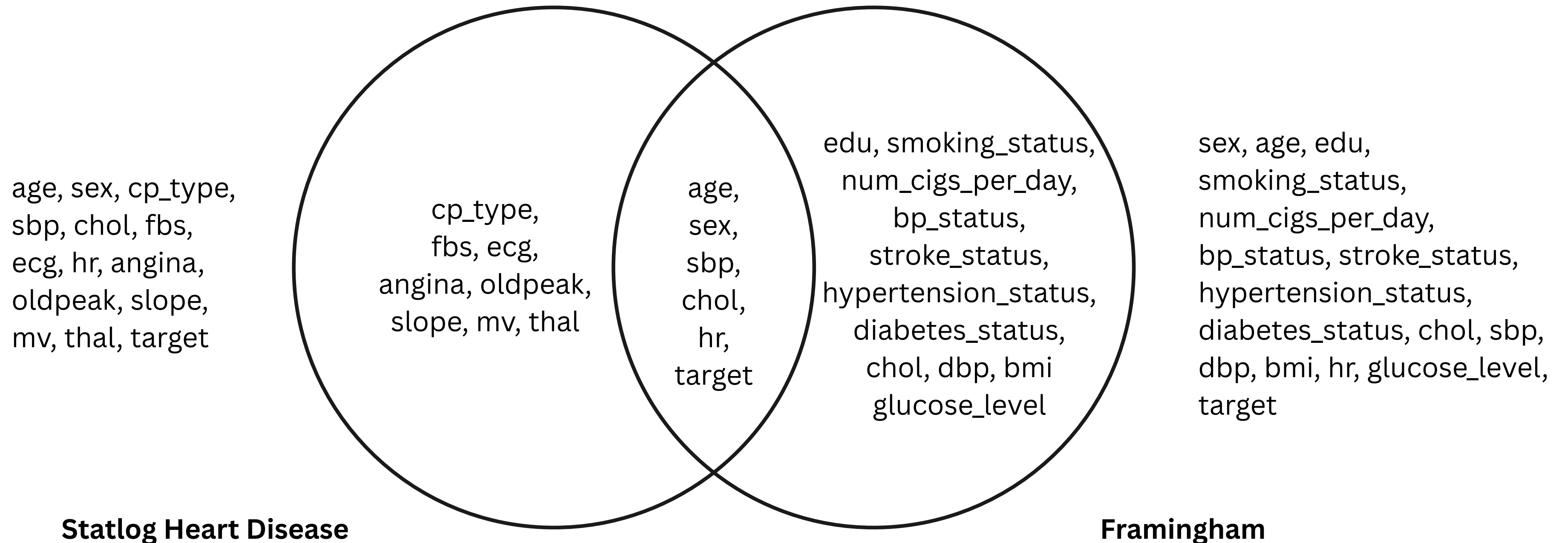
- cp → cp_type
- trestbps → sbp
- restecg → ecg
- thalach → hr
- exang → angina
- ca → mv

Stroke

- id -/->
- gender → sex
- hypertension → hypertension_status
- heart_disease → heart_disease_status
- ever_married → marital_status
- Residence_type → residence_type
- avg_glucose_level → glucose_level
- stroke → target

4. Concatenation Setup

Example of 2-Dataset Concatenation



4. Concatenation Setup

Example of 2-Dataset Concatenation

Statlog-Framingham-Union Columns

age, sex, cp_type, sbp, chol, fbs, ecg, hr, angina, oldpeak, slope, mv, thal, target, edu, smoking_status, num_cigs_per_day, bp_status, stroke_status, hypertension_status, diabetes_status, dbp, bmi, glucose_level

Statlog-Framingham-Intersection Columns

age, sex, sbp, chol, hr, target

5. Concatenation Result

Dataset Combinations

	chd	framingham	heart	statlog	stroke	count
column						
target	✓		✓	✓	✓	5
age	✓		✓	✓	✓	5
sbp	✓		✓	✓		4
sex			✓	✓	✓	4
bmi	✓		✓		✓	3
chol			✓	✓		3
hr			✓	✓		3

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5. Concatenation Result

	chd	framingham	heart	statlog	stroke	count
column						
mv			✓	✓		2
angina			✓	✓		2
cp_type			✓	✓		2
thal			✓	✓		2
ecg			✓	✓		2
oldpeak			✓	✓		2
smoking_status		✓			✓	2
fbs			✓	✓		2
glucose_level		✓			✓	2
slope			✓	✓		2
hypertension_status		✓			✓	2

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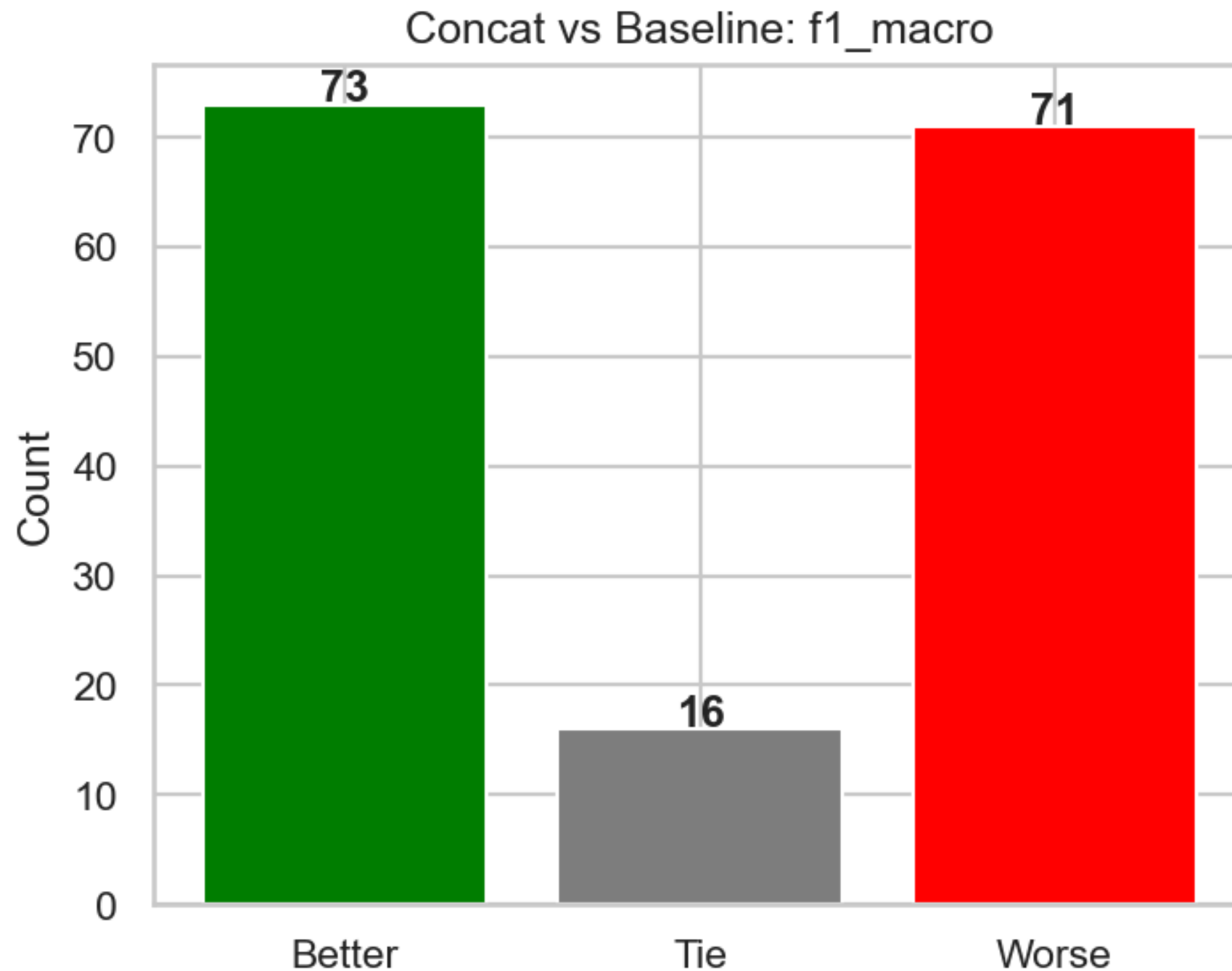
5. Concatenation Result

	chd	framingham	heart	statlog	stroke	count
column						
stroke_status		✓				1
tobacco	✓					1
typea	✓					1
residence_type					✓	1
adiposity	✓					1
num_cigs_per_day		✓				1
marital_status					✓	1
ldl	✓					1
heart_disease_status					✓	1
famhist	✓					1

	chd	framingham	heart	statlog	stroke	count
column						
edu		✓				1
diabetes_status		✓				1
dbp		✓				1
bp_status		✓				1
alcohol	✓					1
work_type					✓	1

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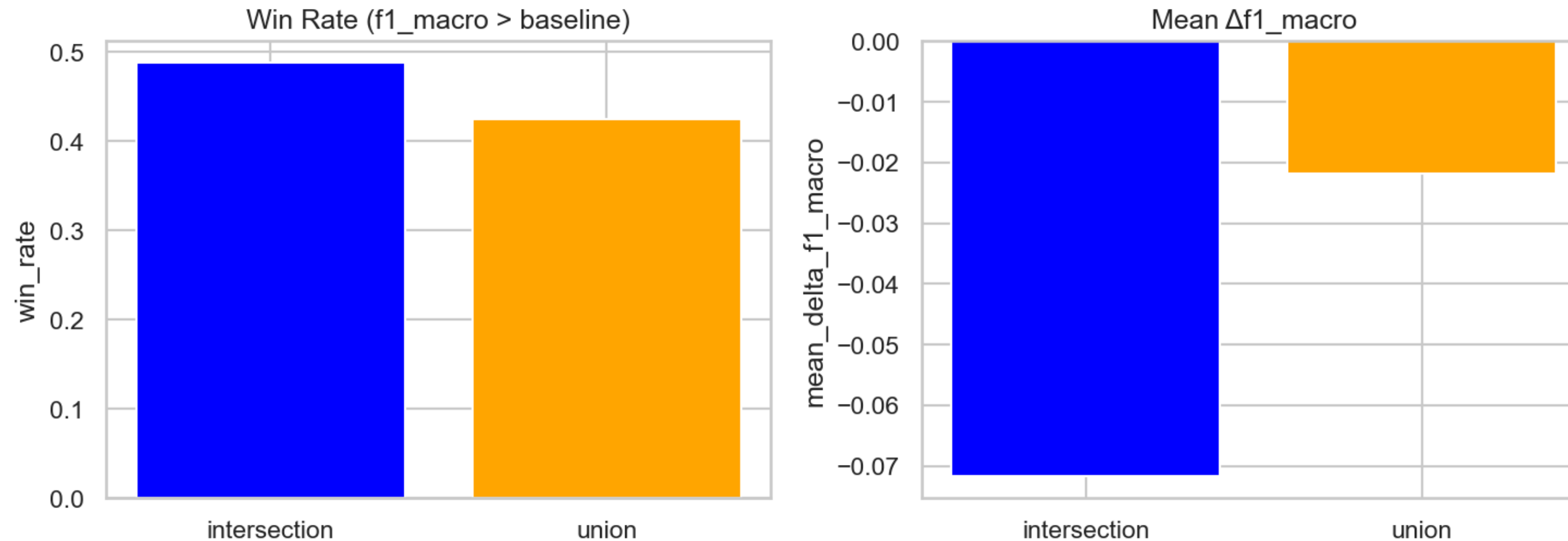
5. Concatenation Result



- Out of 160 comparisons, Concat wins 73, loses 71, ties 16
- Win rate of ~46% – concat does not consistently outperform baseline
- Marginal improvement suggests simple concatenation is insufficient for generalization

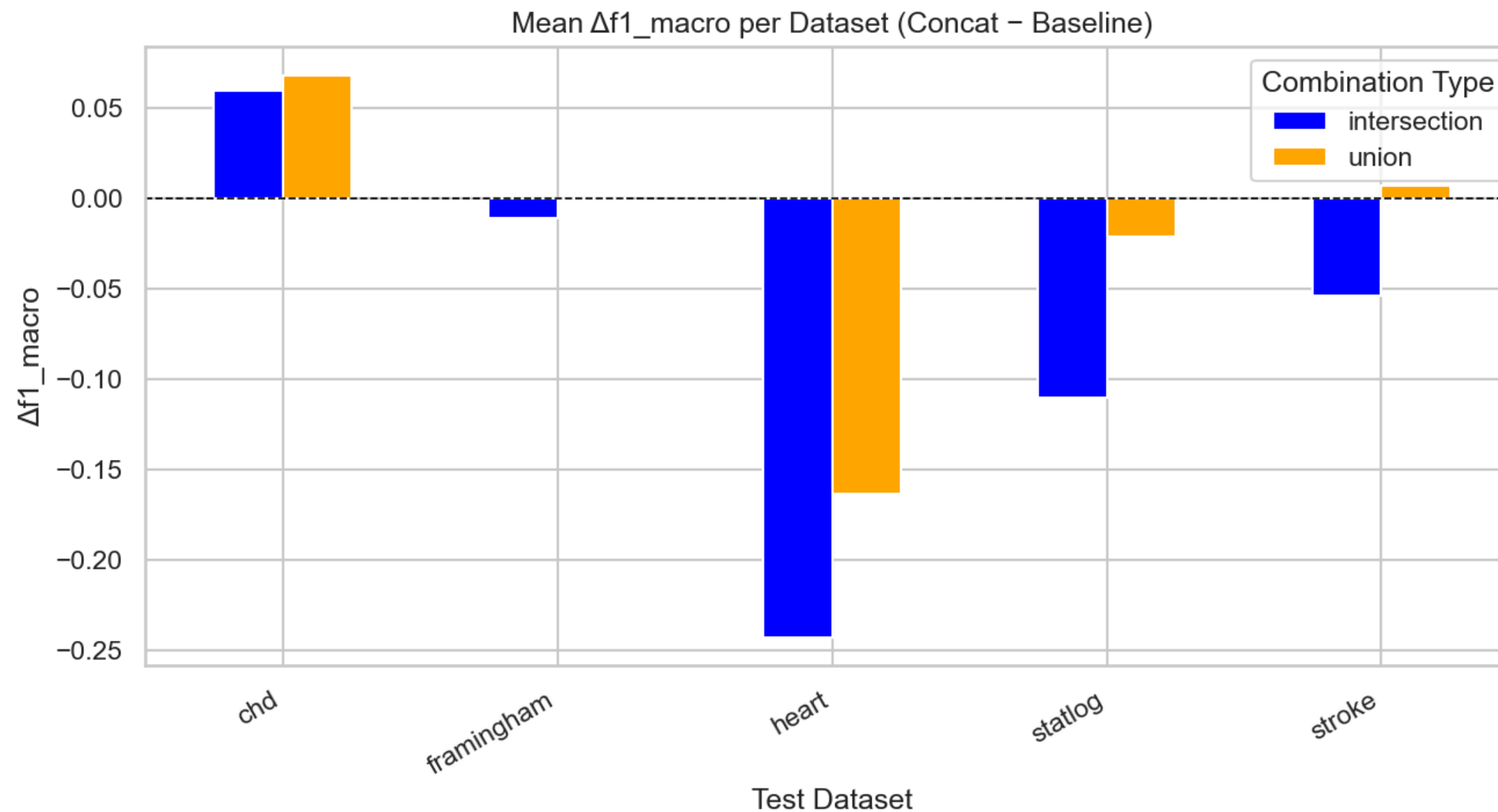
5. Concatenation Result

Union vs Intersection: Generalization Comparison



- Intersection slightly outperforms Union (win rate ~49% vs ~43%)
- However, both strategies show negative mean $\Delta f1_macro$
- Imputing missing values in Union may introduce noise, hurting performance

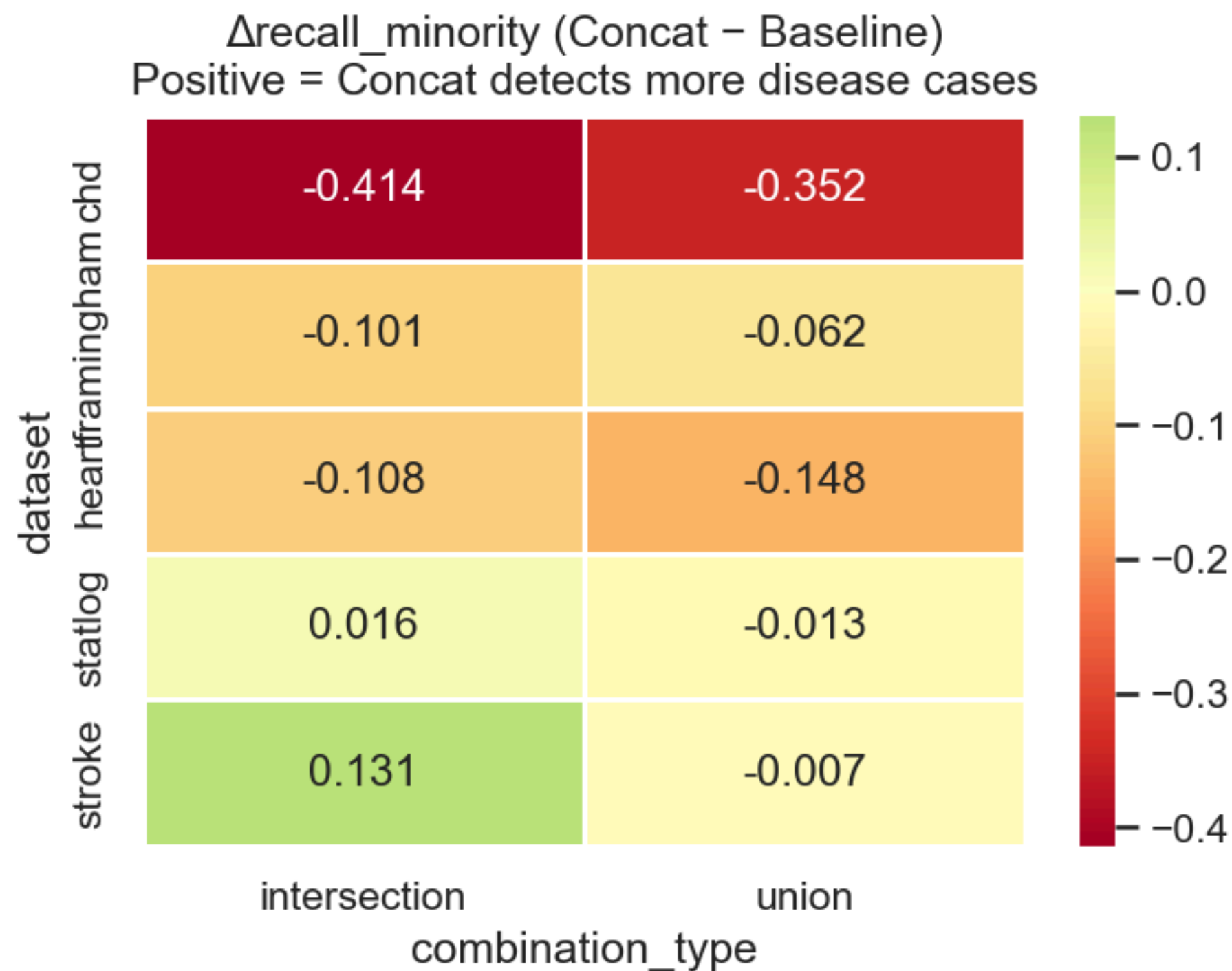
5. Concatenation Result



- **chd** is the only dataset that benefits from concatenation (+0.06)
- **framingham** remains largely unchanged (~0.00)
- **heart**, **statlog**, and **stroke** show notable degradation – heart drops up to -0.25
- Datasets with stronger single-dataset baselines tend to suffer more from concat

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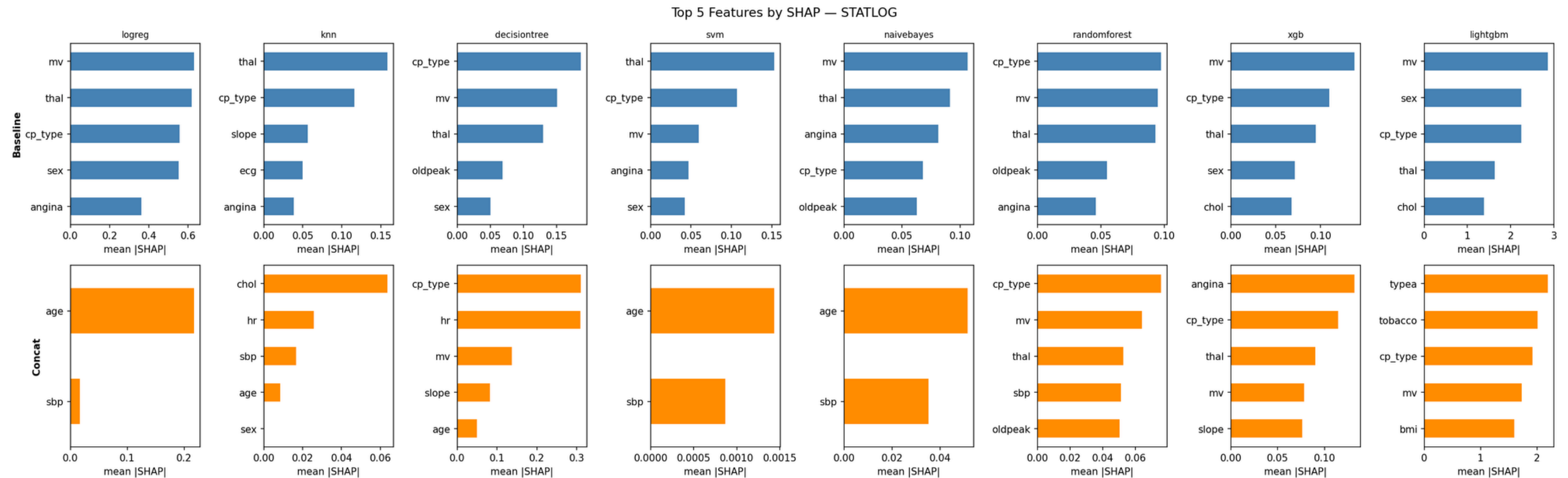
5. Concatenation Result



- Most datasets **show negative Δrecall** – concat detects fewer disease cases than baseline
- **chd** drops most severely (-0.414 intersection, -0.352 union)
- **stroke + intersection** is the only case with improvement (+0.131)

6. Feature Importance

1. Statlog

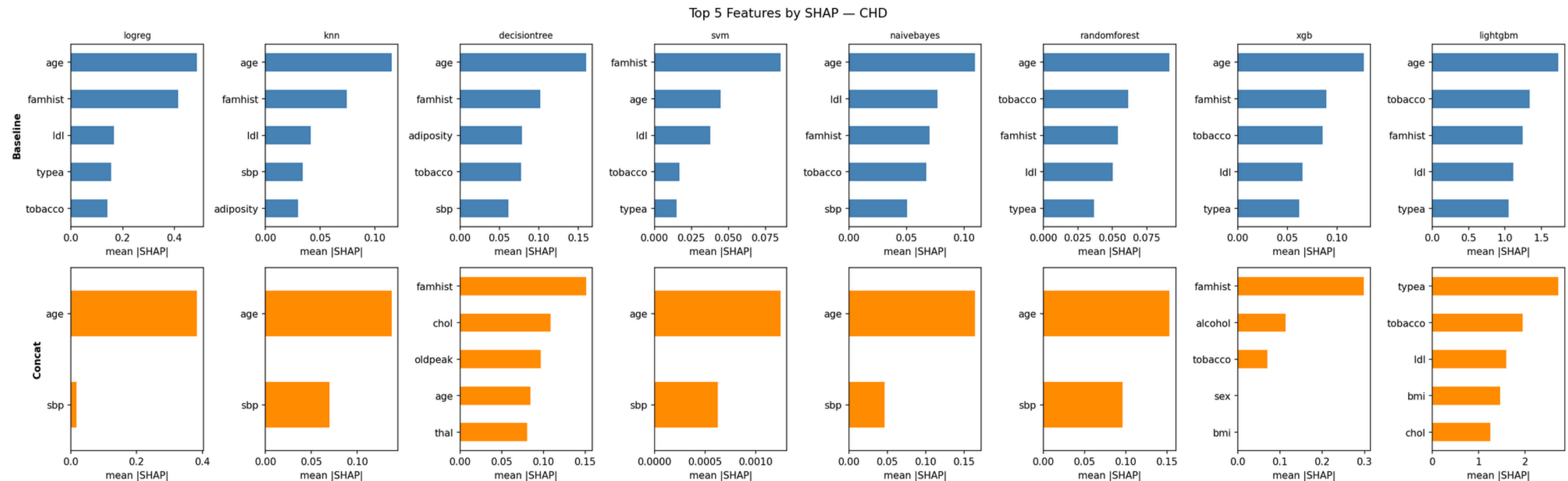


- Baseline: mv, thal, cp_type dominant
- Concat: age becomes dominant, typea and tobacco enter — non-cardiac features override

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6. Feature Importance

2. CHD

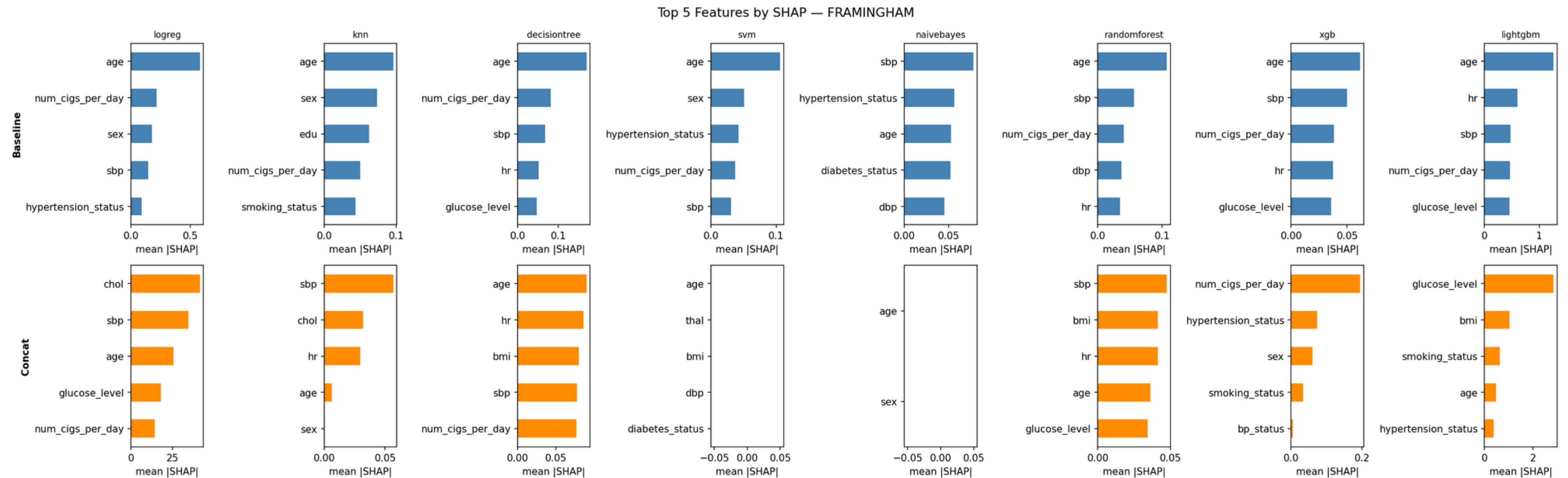


- Baseline: age and famhist are dominant across most models
- Concat: chol, oldpeak, thal enter the top 5 — features from other datasets interfere with CHD-specific signals

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6. Feature Importance

3. Framingham

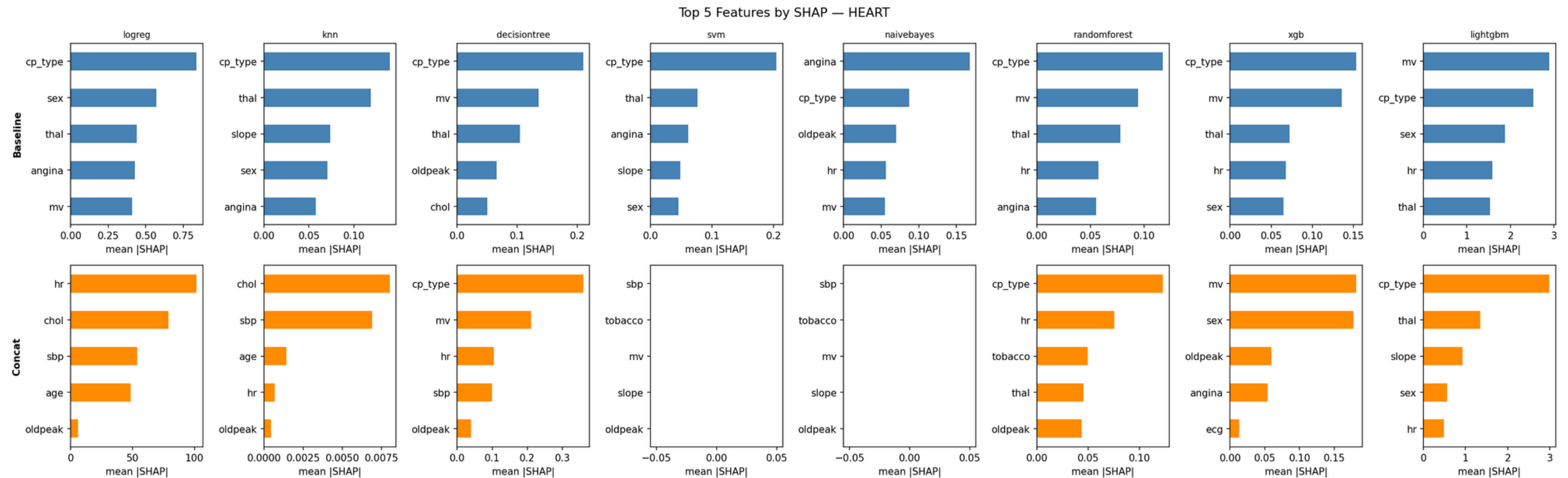


- Baseline: age and sbp consistently dominant
- Concat: chol and glucose_level replace some baseline features — moderate feature shift

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6. Feature Importance

4. Heart

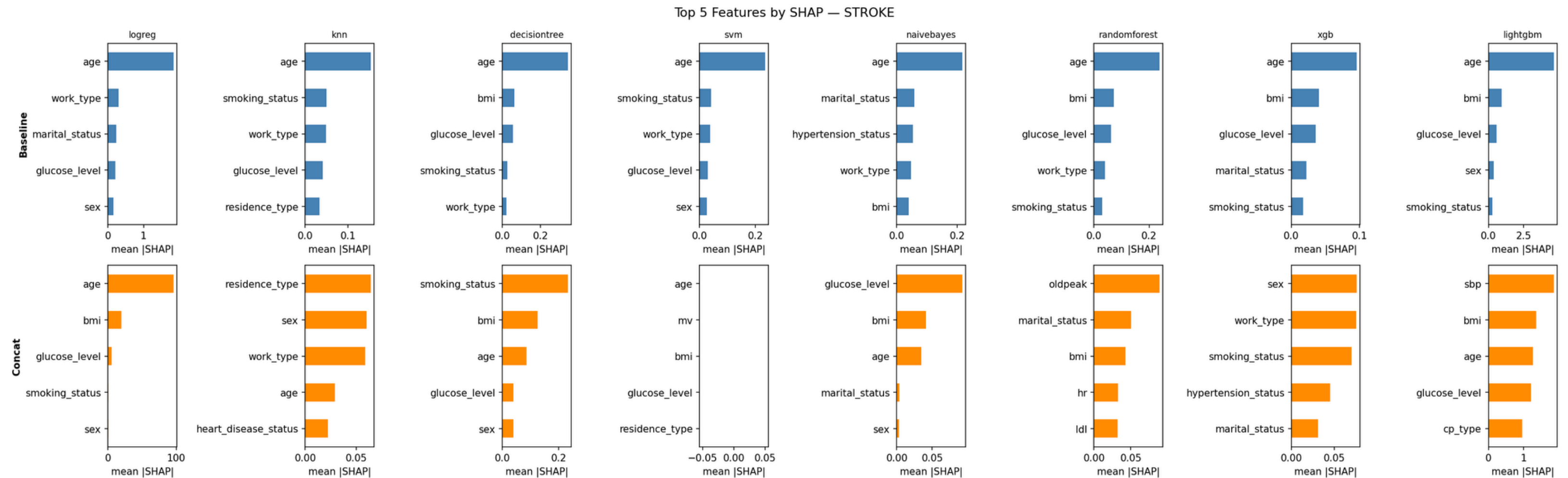


- Baseline: cp_type and mv are strong cardiac-specific signals
- Concat: hr, chol, sbp, tobacco take over — cardiac features are diluted

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6. Feature Importance

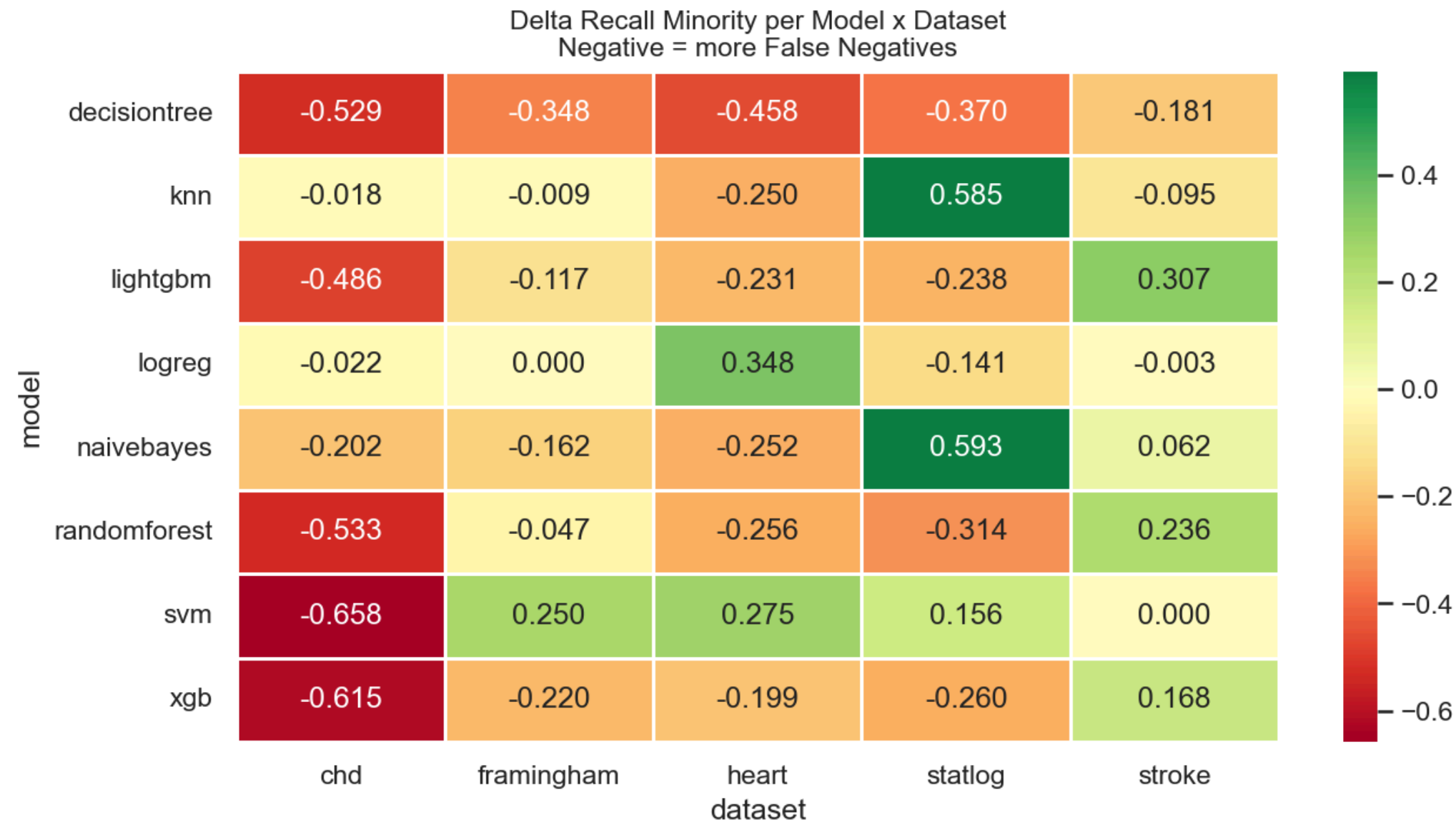
5. Stroke



- Baseline: age dominant across all 8 models — very consistent signal
- Concat: feature importance spreads to oldpeak, ldl, sbp, cp_type — cardiovascular features from other datasets enters

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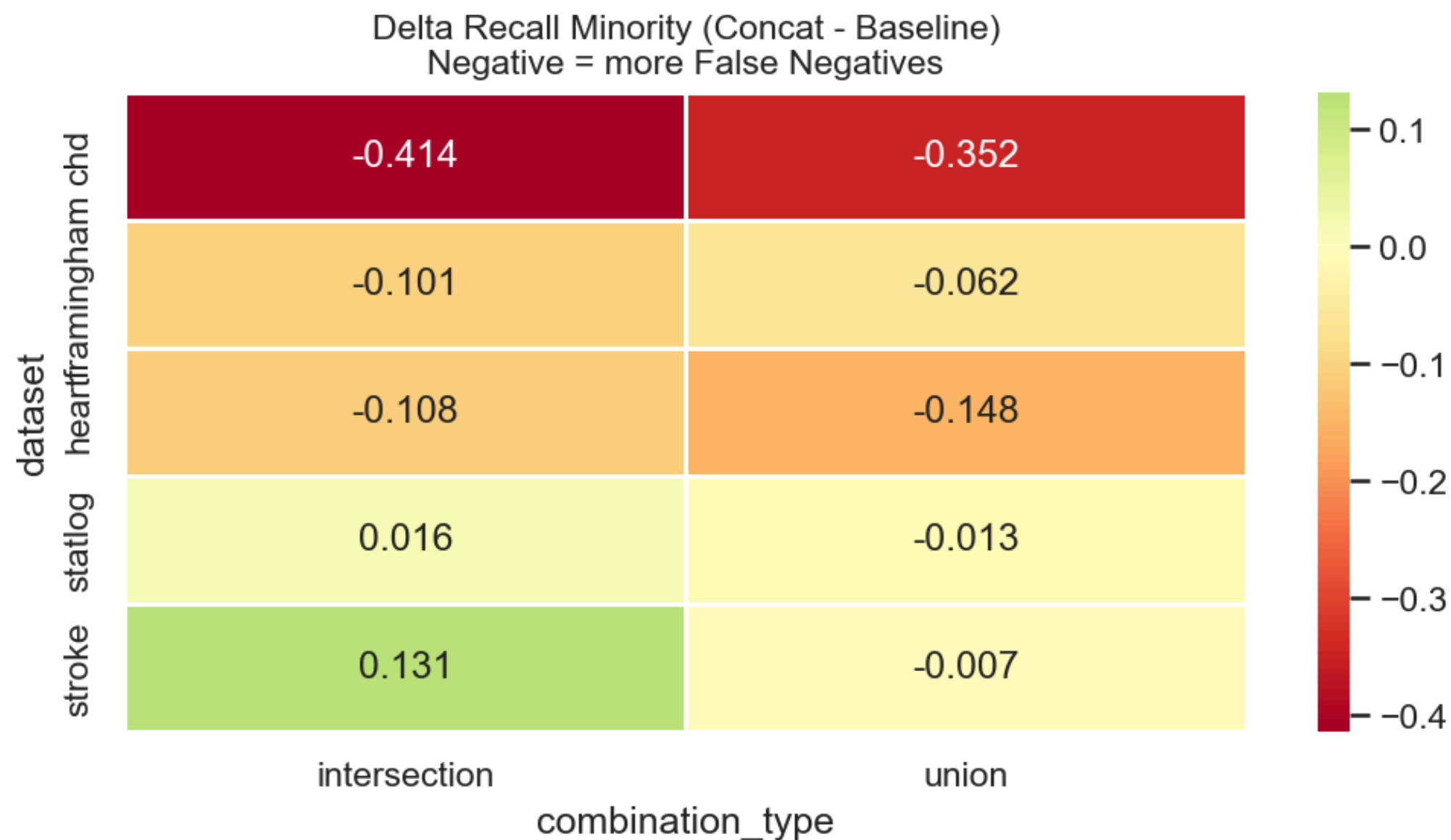
7. Error Analysis



- **Decision Tree** degrades across all 5 datasets – most consistently harmful model
- **SVM** and **XGBoost** drop severely on chd (-0.658, -0.615) but recover on other datasets
- **KNN** and **Naive Bayes** benefit on statlog (+0.585, +0.593) – simple models adapt better
- **LogReg** improves on **heart** (+0.348) but hurts on **statlog** (-0.141)
- No single model consistently benefits from concat across all datasets

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7. Error Analysis



- **chd** suffers the most — recall drops by -0.414 (intersection) and -0.352 (union)
- **stroke intersection** is the only case with meaningful improvement ($+0.131$)
- **statlog and framingham** show minor degradation overall
- Union consistently performs slightly better than intersection on recall minority

8. Conclusion

1. Does concatenation improve generalization?

- No clear improvement — win rate only 46% (73 wins, 71 losses out of 160 comparisons)
- Both Union and Intersection show negative mean $\Delta f1_macro$ overall

2. Which datasets benefit?

- chd — the only dataset that consistently improves with concat
- framingham — largely unchanged
- heart, statlog, stroke — degrade, with heart dropping most severely (up to -0.25)

3. Union vs Intersection?

- Intersection slightly better on win rate (~49% vs ~43%)
- Union introduces noise via imputation, hurting performance on feature-rich datasets

4. Clinical concern (Recall Minority)

- Most datasets detect fewer disease cases after concat
- chd drops most severely (-0.414)
- stroke intersection is the only meaningful improvement (+0.131)

8. Conclusion

5. Why does concat fail?

- Different feature distributions across datasets confuse the model
- Imputed missing values (Union) add noise
- Stronger single-dataset baselines are hard to beat

Limitation

- Only 5 datasets with notably different characteristics — each focuses on different clinical aspects of cardiovascular disease
- No hyperparameter tuning per concat configuration

Future Work

- Expand to more diverse and larger datasets
- Tune hyperparameters per concat configuration for fairer optimization
- Change ways to handle class imbalance

Source Code

<https://github.com/PATRECOG-RaoTamMakKwaPeudSSR/multidataset-health>